UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 004 · Implicit functions, problems on implicit and composite functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9 · S003·26	Deck 004 / 003, slide number (rendered .pptx)
T004a 00:13:09	Transcript 004 part I: "Implicit functions and problems"
T004b 00:02:31	Transcript 004 part II: "Problems on composite and implicit functions"
T003b 00:10:33	Transcript 003 part II (implicit problems at its end)
CW U1 P17 · ESA Jan-24 s3 1d	Unit 1 class-work problem 17 · past-paper question

Contents

1. 1 · Implicit functions, level surfaces and the differential
2. 2 · Implicit differentiation in two variables
3. 3 · Implicit functions of three variables
4. 4 · Composite functions and the chain rule
5. 5 · Composite functions in polar coordinates
6. Past-paper questions

1 Implicit functions, level surfaces and the differential

Definitions

Implicit function. A function given by a relation $f(x, y) = c$ (for example $x^3y + 3xy^2 = 3$) rather than by a formula $y = g(x)$. S004·3 · T004a 00:00:18

Level surface (level curve). The set where $z = f(x, y) = c$ (or $f(x_1, \ldots, x_n) = c$). Examples: equipotential surfaces, isotherms. The cylinders $x^2 + y^2 = c$ and the spheres $x^2 + y^2 + z^2 = c$ are layers, one for each c. S004·4 · T004a 00:02:35 · T004a 00:03:29

Total differential. For $z = f(x, y)$: $dz = f_x\,dx + f_y\,dy$. For $z = f(x_1, \ldots, x_n)$: $dz = f_{x_1}dx_1 + \cdots + f_{x_n}dx_n$. The total derivative is also called the differential. S004·3 · S004·4 · T004a 00:02:04

Statements in the class material

> *"The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation."* S004·4

> *"If z represents a level surface, i.e. if $z = f(x, y) = c$, then $dz = 0$. The total derivative is also called the differential."* S004·4

Key points

1. Integration by substitution uses the differential: with $t = x^2$, $dt = 2x\,dx$, so $\int e^{x^2} x\,dx = \frac{1}{2}\int e^t\,dt$. ESA ★ (transcript only) T004a 00:07:57 · T004a 00:08:29
2. Errors and approximations, another application of the differential, come at the end of the unit. ESA ★ (transcript only) T004a 00:08:47
3. Restriction to remember: the formula needs $z = f(x, y)$ to be a constant (a level curve). Move everything to one side first: $f(x, y) = c$ or $f(x, y) = 0$. ESA ★★★ T004a 00:01:49 · T004a 00:02:04
4. Level surfaces in physics: equipotential surfaces (constant potential) and isotherms (constant temperature). ESA ★ T004a 00:02:35
5. $x^2 + y^2 = c$ gives a family of cylinders (layers), and $x^2 + y^2 + z^2 = c_1$ a family of spheres about the origin: $c_1 = 0$ is the origin, $c_1 = 1$ the unit sphere, larger c_1 larger spheres. ESA ★ S004·4 · T004a 00:03:10 · T004a 00:03:42
6. $f(x_1, \ldots, x_n) = c$ is still a level surface for $n > 3$, even though it cannot be plotted; its total differential is still 0. ESA ★ T004a 00:04:34

2 Implicit differentiation in two variables

Statements in the class material

> *"Setting* $\dfrac{dz}{dx} = 0$ *leads to a simple computation formula for implicit differentiation:* $f_x \cdot 1 + f_y \dfrac{dy}{dx} = 0$*."* S004·5

Key points

1. Differentiating an implicit relation directly needs the product rule on every term, and the right-hand constant gives 0. ESA ★★ S004·3 · T004a 00:00:48
2. The formula $dy/dx = -f_x/f_y$ gives the same answer and is far easier when f is complicated. ESA ★★ T004a 00:06:19 · T004a 00:06:36
3. There are two ways to solve "find dy/dx": differentiate both sides directly, or use $-f_x/f_y$. Both are accepted. ESA ★★ S003·27 · T003b 00:10:33
4. The slope dy/dx at a point gives the tangent line $y - y_0 = m(x - x_0)$. ESA ★★ S004·7 · T004a 00:12:01
5. For $(\cos x)^y = (\sin y)^x$ the result needs $\cos x > 0$ and $\sin y > 0$, because of the logarithms. ESA ★ T004a 00:18:12
6. Implicit and composite together: if $u = \tan(x^2 + y^2)$ and $x^2 - y^2 = 2$, use the relation to eliminate y (or find dy/dx) before differentiating. ESA ★★ CW U1 P16 · CW U1 P18

Why?

Why is $dy/dx = -f_x/f_y$?
Along the curve $f(x, y) = c$ the value of f does not change, so $df = f_x\,dx + f_y\,dy = 0$. Divide by dx: $f_x + f_y \dfrac{dy}{dx} = 0$. The branch diagram (Figure G2.1) shows the same sum.
S004·4 · S004·5 · T004a 00:07:39

Why is the level-curve slope $-f_x/f_y$ and not $-f_y/f_x$ (as printed on slide 004·4)?
From $f_x + f_y\,y' = 0$ you get $y' = -f_x/f_y$. Test it on $x^2 + y^2 = 5$ at $(2, 1)$: $-f_x/f_y = -2x/(2y) = -2$, which matches differentiating directly; $-f_y/f_x$ would give $-1/2$.
ADDED FOR CLARITY S004·4 · S004·5

Exam traps

Exam trap. Slide 004·4 prints $dy/dx = -f_y(x, y)/f_x(x, y)$. The correct formula is $dy/dx = -f_x/f_y$. ADDED FOR CLARITY S004·4

Exam trap. In $(\cos x)^y$, with respect to x it is a power ($y(\cos x)^{y-1}(-\sin x)$); with respect to y it is an exponential ($(\cos x)^y \ln \cos x$). S004·8 · T004a 00:16:02

3 Implicit functions of three variables

Definitions

Implicit function of three variables. $f(x, y, z) = 0$ with z (or y) regarded as a function of the remaining variables; the other variables are intermediate variables. S004·6 · T004a 00:09:05

Statements in the class material

> *"Let $f(x, y, z) = 0$ be the equation of an implicit function of three variables x, y, z. Suppose y and z are functions of x, then f is a function of one independent variable x and y, z are intermediate variables."* S004·6

Key points

1. For $f(x, y, z) = 0$, x and y are independent, so dz alone has no meaning; find the partials z_x, z_y. ESA ★★ T004a 00:09:38
2. For $\partial z/\partial x$ treat y as constant; for $\partial y/\partial x$ treat z as constant. ESA ★★ S004·6 · S004·9 · T004a 00:19:00
3. The formula avoids separating z from a messy equation such as $x^3y^2 + 3xyz + \sin(e^z) = c$. ESA ★ T004a 00:10:55
4. Check that the given point satisfies $f = 0$ before substituting. ESA ★★ (added for clarity) S004·9 · T004a 00:18:39

Exam traps

Exam trap. The lecture works problem 3 at $(1, 0, 2)$ and gets 0 for both derivatives, but $(1, 0, 2)$ is not on $z^3 + xy - y^2z = 6$ (it gives 8). Use the slide's point $(0, 1, 2)$: $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$. (added for clarity) S004·9 · T004a 00:20:47

Exam trap. Do not differentiate $-f_x/f_z$ as if z were a constant when you need second derivatives: z is still a function of x and y (W3.3). (added for clarity) ESA Jan-24 s3 1d

4 Composite functions and the chain rule

Definitions

Chain rule (several variables). If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables u, v, then $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$ and $\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. S003·13 · T003a 00:21:07

The chain of a problem. The list of dependencies, e.g. "u depends on x, y; x, y depend on r, t; so u is a composite function of r, t". Knowing it fixes which chain-rule formula to write. T004b 00:02:31

Statements in the class material

> *"We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v."* S003·13

Key points

1. It is essential to know the chain given in the problem: what depends on what. The formula follows from it. ESA ★★★ T004b 00:02:31
2. When u is a composite function of r, θ (or s, t), eliminate x and y from the final answer: you must know why you substitute. ESA ★★★ T004b 00:07:52 · T004b 00:10:28

Why?

Why does $u = f(x - y,\ y - z,\ z - x)$ give $u_x + u_y + u_z = 0$?
Each argument is a difference whose derivatives sum to zero: for $r = x - y$, $r_x + r_y + r_z = 1 - 1 + 0 = 0$. Adding the three chain-rule expressions gives $f_r(0) + f_s(0) + f_t(0)$. In words: moving x, y, z by the same amount changes none of the arguments.

ADDED FOR CLARITY ESA Jan-24 s2 1d

Why do the chain-rule answers keep f_r and f_θ?
The outer function f is unknown, so its partial derivatives cannot be computed; only the inner derivatives ($r_s = t$, $\theta_s = s/\sqrt{s^2 + t^2}$, ...) can.

S003·17 · T003a 00:26:28

5 Composite functions in polar coordinates

Key points

1. Functions of $x^2 + y^2$ alone (e.g. $\sin(x^2 + y^2)$, $e^{x^2+y^2}$) have $\partial u/\partial\theta = 0$ in polar coordinates, since $x^2 + y^2 = r^2$ does not involve θ. ESA ★★ ADDED FOR CLARITY S004·10 · S004·13 · S004·14
2. $\tan^{-1}(y/x)$ is exactly θ in polar coordinates, so $\partial u/\partial\theta = 1$. ESA ★★ ADDED FOR CLARITY S004·15

Exam traps

Exam trap. Do not stop with x and y still in an answer that should be in r, θ or s, t. T004b 00:07:52

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. In scope: one implicit-function question (Jan 2024 set 3 Q1.d, 6 marks) and five composite-function (chain rule) questions (Jan 2024 set 1 Q1.d and set 2 Q1.d, 5 marks; set 3 Q1.b, set 4 Q1.b and June 2024 Q1.a, 6 marks). Polar-coordinate problems and tangent lines to implicit curves were not asked in these papers; they are computed in class and in CW problems 15–18. Total-derivative and rate problems (class 003) are left out.

3 · Implicit functions of three variables

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.d · solved in W3.3

For the point on the surface $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -[x \log(ex)]^{-1}$.

4 · Composite functions and the chain rule

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q1.d · solved in W4.3

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

ESA Jan 2024 (5 marks) UE23MA141A set 2, Q1.d · solved in W4.4

If $u = f(x - y, y - z, z - x)$, find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.b · solved in W4.5

If $z = f(u, v)$ and $u = x^2 - y^2$, $v = 2xy$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2 + v^2)^{1/2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.b · solved in W4.6

If $z = f(x, y)$ and $x = e^u \sin v$, $y = e^u \cos v$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA June 2024 (6 marks) UE23MA141A Q1.a · solved in W4.7

If $z = \log(u^2 + v)$, where $u = e^{x+y^2}$, $v = x + y^2$, then find the value of $2y\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$.